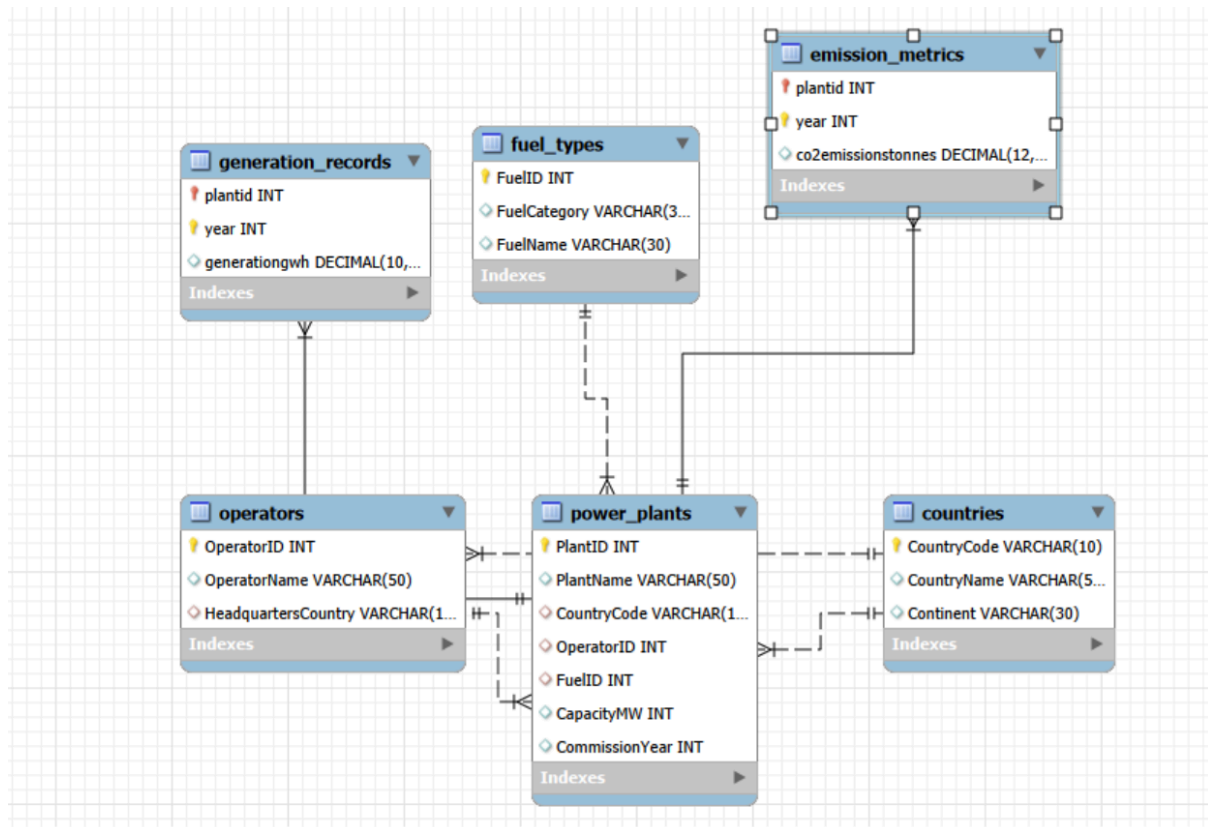


Assignment 3

Krishna Santhosh



Query 1

```

1 • SELECT power_plants.Plantname, countries.Countryname, operators.OperatorName,
2       fuel_types.FuelCategory, fuel_types.FuelName, power_plants.CapacityMW, power_plants.CommissionYear
3 FROM power_plants, countries, operators, fuel_types
4 WHERE power_plants.CountryCode = countries.CountryCode
5 AND power_plants.OperatorID = operators.OperatorID
6 AND power_plants.FuelID = fuel_types.FuelID
7 ORDER BY capacitymw DESC;
  
```

Plantname	Countryname	OperatorName	FuelCategory	FuelName	CapacityMW	CommissionYear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

Query 2

```

10  -- Query 2
11  • SELECT
12      power_plants.PlantName, power_plants.CountryCode, generation_records.year, generation_records.generationgwh
13  FROM power_plants, generation_records
14  WHERE
15      power_plants.plantid = generation_records.plantid
16  AND generation_records.year = 2024
17  ORDER BY generation_records.generationgwh DESC;

```

PlantName	CountryCode	year	generationgwh
Three Gorges	CHN	2024	98800
Itaipu	BRA	2024	89000.1
Tuoketuo	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariwa	JPN	2024	0

Query 3

```

20  -- Query 3
21  • SELECT
22      power_plants.PlantName, power_plants.CountryCode, generation_records.year,
23      generation_records.generationgwh, emission_metrics.co2emissionstonnes
24  FROM power_plants, generation_records, emission_metrics
25  WHERE
26      power_plants.plantid = generation_records.plantid
27  AND power_plants.PlantID = emission_metrics.plantid
28  AND generation_records.year = 2024
29  ORDER by emission_metrics.co2emissionstonnes ASC;

```

PlantName	CountryCode	year	generationgwh	co2emissionstonnes
Grand Coulee	USA	2024	21500.5	0
Itaipu	BRA	2024	89000.1	0
Kashiwazaki-Kariwa	JPN	2024	0	0
Three Gorges	CHN	2024	98800	0
Loy Yang	AUS	2024	14200.4	1950000
Janschwalde	DEU	2024	18400.2	2450000
Vindhyachal	IND	2024	35600.8	3100000
Tuoketuo	CHN	2024	71000.3	5200000

Query 4

```

32  -- Query 4
33  WITH total as(
34      SELECT plantID, sum(generationgwh) AS total_generation
35      FROM generation_records
36      group by plantID
37  )
38
39  SELECT operators.OperatorName, operators.HeadquartersCountry, SUM(total.total_generation) AS cumulativepower
40  FROM power_plants, operators, total
41  WHERE total.plantID = power_plants.PlantID
42  AND power_plants.OperatorID = operators.OperatorID
43  GROUP BY operators.OperatorName, operators.HeadquartersCountry
44  ORDER BY cumulativepower DESC;

```

OperatorName	HeadquartersCountry	cumulativepower
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

Query 5

```

48  WITH powertotal AS (
49      SELECT power_plants.CountryCode, SUM(generation_records.generationgwh) AS totalgeneration
50      FROM power_plants, generation_records
51      WHERE power_plants.PlantID = generation_records.plantid
52      GROUP BY power_plants.CountryCode
53  ),
54  emission_totals AS (
55      SELECT power_plants.CountryCode, SUM(emission_metrics.co2emissionstonnes) AS totalemissions
56      FROM power_plants, emission_metrics
57      WHERE power_plants.PlantID = emission_metrics.plantid
58      GROUP BY power_plants.CountryCode
59  )
60
61  SELECT countries.CountryName, powertotal.totalgeneration, emission_totals.totalemissions
62  FROM countries, powertotal, emission_totals
63  WHERE countries.CountryCode = powertotal.CountryCode
64  AND countries.CountryCode = emission_totals.CountryCode

```

CountryName	totalgeneration	totalemissions
China	169800.3	5200000
Brazil	89000.1	0
India	35600.8	3100000
United States	21500.5	0
Germany	18400.2	2450000
Australia	14200.4	1950000
Japan	0	0